

SPATIAL AFFORDANCE TOKENIZATION VS. PASSIVE CLASSIFICATION

Lifting 2D Pixels into 3D Metric Affordance Tokens for Action Decoders and World Models

RAW OBSERVATION

RGB-D Stream

Calibrated RGB + Depth
640 × 480 × 4

DMA Coherent Buffer

$$I_t \in \mathbb{R}^{H \times W \times 4}$$

SPATIAL ENCODER

ViT / DINOv2 Backbone

14 × 14 Patch Extraction
Multi-Head Self-Attention

Dense Feature Extraction

$$Z_{\text{vis}} \in \mathbb{R}^{N \times D_{\text{model}}}$$

❌ PASSIVE CLASSIFICATION (Digital ML Paradigm)

- Global Average Pooling over spatial patch grid
- Emits categorical probability vector: $p(\text{"ceramic mug"}) = 0.94$
- **Physical AI Failure:** Contains zero metric coordinates, zero contact normals, zero grasp reachability, and zero collision boundaries. Unusable by motor controllers.

✅ 3D SPATIAL AFFORDANCE TOKENS (Physical AI Paradigm)

- 3D Metric Unprojection: $P_{3D} = T_C^W \cdot K^{-1}[u, v, 1]^T \cdot d(u, v)$
- Metric Grasp Frame: $T_{\text{grasp}} = [p_{\text{target}}, q_{\text{quat}}] \in SE(3)$
- Surface Normal & Friction: $\hat{n} \in S^2, \quad \hat{\mu}_{\text{friction}} = 0.45 \pm 0.05$
- Obstacle Clearance Boundary: $r_{\text{clearance}} = 0.08 \text{ m (for CBF Invariant)}$

→ **DOWNSTREAM ACTION CONSUMPTION:** Feeds directly into **Station 4** (Latent World Models), **Station 5** (1 Hz VLM Intent Leases), and **Station 6** (20 Hz Action Chunking Decoders)